A Lightweight and Interpretable Approach for Stress Detection in Social Media using Feature Fusion and Machine Learning

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***Abstract*— Stress detection from social media is becoming increasingly important for understanding mental health, as people often express their emotions and experiences online. While many existing approaches rely on advanced deep learning models like BERT, these methods are often resource-intensive, complex to implement, and difficult to interpret. This study proposes a simpler and more efficient approach using Logistic Regression with TF-IDF features on the Dreaddit dataset. The model not only analyzes textual data but also incorporates behavioral information, such as word count, through feature fusion. To improve interpretability, it highlights key words that contribute to stress prediction. Experimental results show that the proposed approach achieves good performance while remaining computationally efficient, easy to understand, and suitable for real-world and educational applications.**

List of Acronyms

Acronym	***Meaning***
NLP	*Natural Language Processing*
TF-IDF	*Term Frequency – Inverse Document Frequency*
ML	*Machine Learning*
LR	*Logistic Regression*
EDA	*Exploratory Data Analysis*
XAI	*Explainable Artificial Intelligence*

Keywords*— *Stress Detection, Machine Learning, TF-IDF, Logistic Regression, Feature Fusion, Explainable AI, NLP.

I. INTRODUCTION

The rapid expansion of social media platforms has resulted in a vast amount of user-generated textual data, where individuals regularly share their thoughts, emotions, and mental states. Among various mental health challenges, stress has become a significant concern affecting people across diverse age groups and social contexts. The early detection of stress through the analysis of online content can play an important role in enabling timely support and reducing the risk of more serious psychological conditions.

In recent years, developments in Natural Language Processing (NLP) and machine learning have made it possible to automatically analyze textual data for mental health assessment. A number of studies have focused on stress detection using advanced deep learning techniques, particularly transformer-based models such as BERT, which are effective in capturing contextual relationships within text and often deliver high predictive performance. However, these approaches typically require considerable computational resources, large datasets, and complex training processes, which can limit their usability in resource-constrained or educational environments.

To address these limitations, there is an increasing interest in approaches that are not only efficient but also easier to interpret. Traditional machine learning methods, such as Logistic Regression combined with TF-IDF feature representation, provide a practical alternative due to their lower computational requirements and straightforward implementation. In addition, incorporating simple behavioral features, such as patterns in word usage, can further enhance the model's ability to identify stress-related signals.

In this study, a lightweight and interpretable framework for stress detection is proposed using the Dreaddit dataset. The approach combines textual features with a behavioral attribute, specifically word count, through feature fusion. To improve transparency, the model also identifies key words that contribute to stress prediction. Furthermore, the study explores how stress is distributed across different Reddit communities to gain contextual insights. The main objective is to demonstrate that a simple and efficient model can achieve reliable performance while remaining interpretable and suitable for real-world applications.

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II. LITERATURE REVIEW

Recent advancements in machine learning and natural language processing (NLP) have played a significant role in enabling the analysis of mental health through textual data shared on social media platforms. As individuals increasingly express their emotions and experiences online, these technologies have become valuable tools for understanding psychological well-being.

A systematic review titled *Machine Learning and Natural Language Processing in Mental Health: Systematic Review (2021)* examined multiple studies in this domain and found that most research relies on either clinical datasets or social media data. The review also observed that many existing models tend to validate predefined clinical assumptions rather than uncovering new insights. In addition, it highlighted important challenges such as language dependency and ethical concerns, particularly when applying these models in real-world scenarios.

In recent years, transformer-based models like BERT have demonstrated strong performance in stress detection tasks. For instance, the study *Contextual Stress Detection in Social Media Using Transformer-Based Language Models (2025)* introduced a contextual approach using BERT and showed that it significantly outperforms traditional machine learning models such as SVM, Logistic Regression, and Random Forest. The model achieved an F1-score of 0.89, reflecting the effectiveness of contextual embeddings in capturing stress-related patterns. However, despite their high accuracy, such models often require substantial computational resources and offer limited interpretability**.**

Similarly, the study *Machine Learning Driven Mental Stress Detection on Reddit Posts (2023)* compared traditional machine learning techniques with deep learning approaches. The findings indicated that although advanced models like BERT and ELMo deliver superior performance, classical methods still achieve reasonably good results, with an F1-score of 0.76. This suggests that traditional approaches remain practical and relevant, particularly in environments with limited computational resources.

Another study, *Twitter Sentiment Analysis of Covid-19 Using TF-IDF and Logistic Regression (2020)*, demonstrated the effectiveness of combining TF-IDF feature extraction with Logistic Regression for large-scale text classification tasks. The model achieved an accuracy of 94.71%, highlighting that simpler techniques can still perform efficiently when applied appropriately. Furthermore, the study *A Comparative Analysis of Logistic Regression, Random Forest and KNN for Text Classification (2020)* found that Logistic Regression offers competitive performance while maintaining lower computational complexity, making it a suitable choice for applications where efficiency and simplicity are essential.

III. METHODOLOGY

3.1 Dataset

The Dreaddit dataset, consisting of Reddit posts labeled as stress (1) and non-stress (0), was used in this study. The dataset contains textual content along with additional attributes such as subreddit information.

3.2 Data Preprocessing

The dataset was cleaned by removing missing values and applying text preprocessing techniques such as:

- Converting text to lowercase
- Removing URLs
- Removing special characters

Additionally, a new feature called **word_count** was created to represent the number of words in each post.

3.3 Exploratory Data Analysis

Various visualizations were performed to understand the dataset:

- Distribution of post lengths
- Stress vs non-stress proportion (pie chart)
- Word frequency analysis for stress and non-stress posts
- Correlation analysis of features
- Stress distribution across different subreddits

These analyses provided insights into patterns of stress expression in social media.

3.4 Feature Engineering

Textual data was converted into numerical form using TF-IDF vectorization with the top 500 features.

To enhance model performance, **feature fusion** was applied by combining:

- TF-IDF features (textual data)
- Word count (behavioral feature)

The numerical feature was normalized using MinMaxScaler before combining.

3.5 Model Building

The dataset was split into training and testing sets using an 80-20 ratio.

A Logistic Regression model was trained on the combined feature set. This model was chosen due to its simplicity, efficiency, and suitability for binary classification problems.

3.6 Model Evaluation

The model was evaluated using:

- Accuracy score
- Classification report (precision, recall, F1-score)
- Confusion matrix

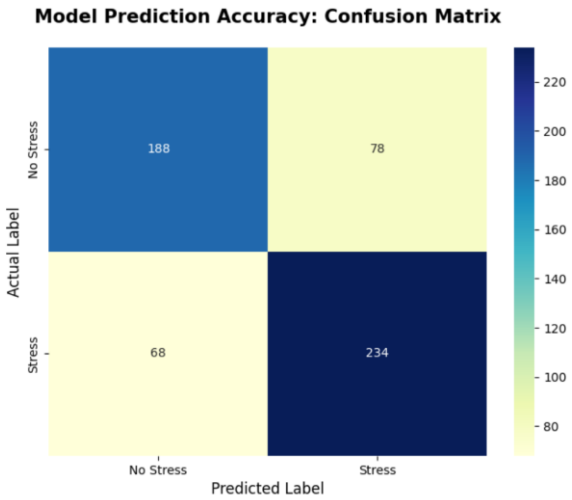
3.7 Explainability

- To improve interpretability, the model's coefficients were analyzed to identify:
- Top words indicating stress
- Top words indicating non-stress
- This helps understand how the model makes decisions.

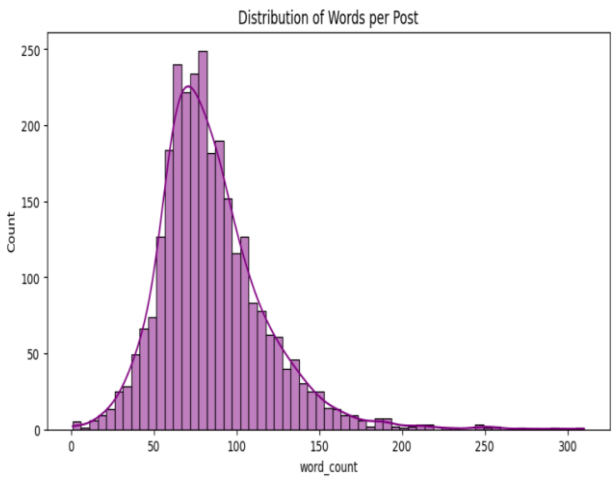
IV. RESULT AND CONCLUSION

The performance of the proposed stress detection model was evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. The Logistic Regression model achieved an accuracy of approximately **76.09%**, indicating that the model is capable of effectively distinguishing between stress and non-stress posts. The classification report further demonstrates balanced performance across both classes, suggesting that the model generalizes reasonably well.

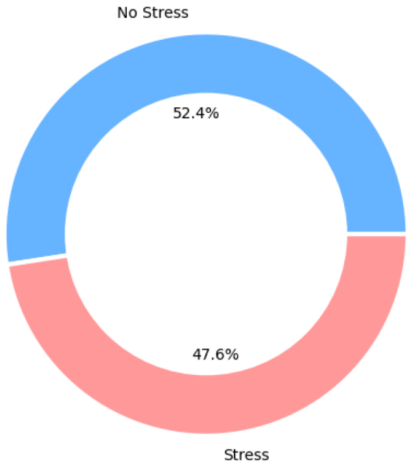
The confusion matrix provides a visual representation of the model's predictions compared to the actual labels. It shows that the model correctly classifies a majority of the instances, with a relatively small number of misclassifications. This indicates that the model is reliable for basic stress detection tasks, although there is still scope for improvement.



Exploratory Data Analysis (EDA) provided useful insights into the dataset and helped in understanding its overall structure. The analysis of post lengths showed that most posts are of moderate length, while a smaller number are longer and more detailed, reflecting deeper user expression. The word count distribution also suggests that individuals experiencing stress may tend to write more descriptive posts.



The class distribution, visualized through pie and bar charts, indicates that the dataset is fairly balanced between stress and non-stress categories. This balance is important, as it helps ensure that the model does not become biased toward any one class during training.



Word frequency analysis revealed noticeable differences in language use. Stress-related posts often contain more emotionally expressive words, whereas non-stress posts tend to include neutral or positive terms. This highlights the role of language patterns in identifying stress.

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Top Stress Words:

   Word  Coefficient
     me     3.299070
     im     2.409539
   cant     2.274409
   even     2.175655
   feel     2.005480
   dont     1.919036
     am     1.874333
anxiety     1.801630
   what     1.726232
fucking     1.666482

Top Non-Stress Words:

      Word  Coefficient
       you    -1.950931
       met    -1.827304
       url    -1.679456
      that    -1.440018
       bit    -1.420066
     years    -1.411869
      many    -1.386144
        be    -1.378563
experience    -1.305608
       guy    -1.273856
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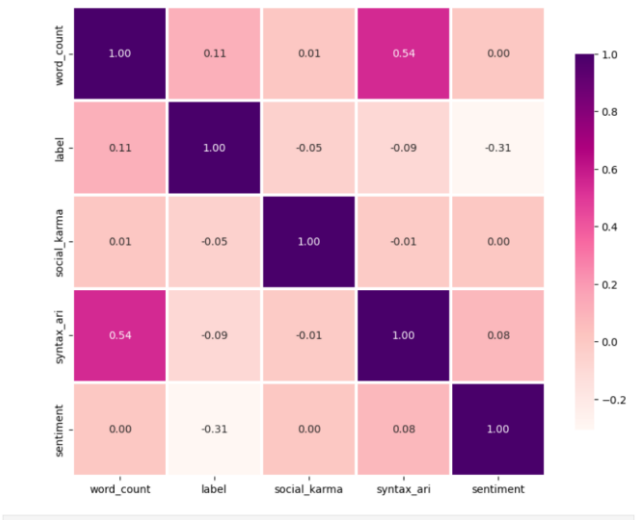
The correlation heatmap offered additional insights into the relationships between features such as word count, sentiment, and readability. While the relationships are not very strong, some features show moderate connections with stress, suggesting they still contribute to prediction.



Subreddit-based analysis further showed that stress levels vary across different communities. Some subreddits have a higher presence of stress-related content, indicating that social context can influence how users express their emotions.

Overall, the results suggest that the proposed lightweight model is capable of capturing meaningful patterns from both textual and behavioral data. Although it may not perform as strongly as more complex deep learning models, it remains an effective, efficient, and interpretable approach for stress detection.

V. CONTRIBUTION

This study presents a simple, efficient, and easy-to-understand approach for detecting stress in social media data using machine learning techniques. The main contributions of the work are summarized as follows:

- A lightweight yet effective stress detection model is developed using TF-IDF feature extraction and Logistic Regression, making it well-suited for environments with limited computational resources.
- The study introduces a feature fusion approach that combines textual features with a behavioral aspect, specifically word count, allowing the model to capture both language patterns and user expression.
- To improve transparency, the model focuses on interpretability by analyzing feature coefficients to identify important words that influence stress and non-stress predictions.
- A comprehensive Exploratory Data Analysis (EDA) is conducted to better understand the dataset, including patterns in post length, word usage, and class distribution.
- The research also examines stress levels across different Reddit communities, offering insights into how social context affects user behavior and emotional expression.
- Overall, the proposed approach strikes a balance between performance, simplicity, and interpretability, providing a practical alternative to complex deep learning models like BERT, which often require significant computational resources.

VI. REFERENCES

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